

Workflow-Based Data Science Pipelines for Digital Phenotyping

Enabling Sustainable & FAIR mobile Health Research

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Abstract

The increasing use of mobile and wearable sensing for digital phenotyping has enabled collection longitudinal datasets across diverse health domains. However, data analysis pipelines remain fragmented and ad hoc, limiting sustainability, and adherence to FAIR principles. Existing workflow systems lack tight integration with study protocols and accessibility for clinical researchers. This paper presents the CARP Data Science Pipeline, an extensible, workflow-based analytics framework integrated with the Copenhagen Research Platform. CARP-DSP enables researchers to define, deploy, and execute human-readable data science workflows spanning back-end and edge environments. The framework emphasizes sustainable research and FAIR compliance, supporting both technical and non-technical users through a domain-specific language and AI-assisted tooling. The system also incorporates mechanisms for cross-system workflow interoperability and privacy-aware analytics.

CCS Concepts

• **Applied computing** → **Health care information systems**; • **Human-centered computing** → **Ubiquitous and mobile computing systems and tools**; • **Information systems** → **Data analytics**; • **Software and its engineering** → **Software design engineering**.

Keywords

mobile sensing, wearable sensing, context-aware computing, mobile health, mHealth, digital phenotyping, sensors

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1 Problem statement

Digital phenotyping is the measurement and inference of health and behaviour through data collected from personal mobile and wearable devices [11, 22]. Using sensor data alongside ecological momentary assessments (EMAs)

[13], researchers have extracted biomarkers [4] for applications in various health domains, including Parkinson's Disease [29], depression [19], bipolar disorder [7], asthma [10], and cardiovascular disease [21].

Despite advances in generalized data collection frameworks, challenges remain in turning raw mobile and wearable data into validated, reproducible insights. The current bottleneck has moved from data collection to data analysis and interpretation [25, 27]. Modern devices generate large volumes of data that require sophisticated pre-processing, feature extraction, and analytical techniques [4, 25].

Current approaches often fail to adequately support the FAIR (Findable, Accessible, Interoperable, and Reusable) principles [28] and sustainable research practices. Data analysis pipelines in digital phenotyping are frequently fragmented, ad hoc, and poorly documented, complicating reproducibility and data sharing between studies and institutions [15].

Another challenge in the digital phenotyping process is the divide between clinicians and technical researchers. Most existing pipelines require advanced programming skills, limiting clinical domain experts from interpreting and designing digital phenotyping pipelines directly [9]. This technical barrier also means that domain experts often lack the knowledge of packaging the resultant software artifacts in shareable and reusable ways [12].

This research addresses these issues by developing a workflow based analytics framework, as an extension to the already existing Copenhagen Research Platform (CARP) platform [3]. Tentatively titled the CARP Data Science Pipeline (CARP-DSP), it is designed to support FAIR-compliant, sustainable, and reproducible digital phenotyping. By building upon the modular and extensible architecture of the CARP, CARP-DSP provides structured, high-level workflow definitions accessible to both technical and non-technical researchers via a domain-specific language that allows the researchers to easily describe their analysis process. In addition, we will also build on the work done in using artificial intelligence and chatbots to [14] in making the low-code environment more accessible to non-technical domain experts.

2 Related Work

2.1 Platforms for Data Collection

Several established platforms like AWARE [8], Beiwe [16], and CARP [2, 3] effectively support mobile sensor data collection, platform data, ecological momentary assessments (EMA), and experience sampling (ESM). AWARE is notable for its extensibility, sensor coverage, and ease of study configuration. However, it requires significant technical expertise for deployment and management and lacks robust cross-platform support [8, 24]. Beiwe emphasizes



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privacy and security, providing strong back-end support yet still demanding considerable technical skills for customization or extension [16]. CARP differentiates itself through modular architecture and scalability, particularly suited for clinical research contexts, but can be challenging for non-technical researchers to configure without additional tooling [2, 3]. However, it boasts strong cross-platform support with both official Android and iPhone applications, a web interface for deploying studies and managing participants, and integration with several wearable devices. Additional relevant platforms include RADAR-base [18] and Purple Robot [20]. Despite these frameworks, many researchers continue developing custom applications to address specialized data collection needs, which continues fragmentation and difficulties in standardizing methodologies across research studies [25, 27].

2.2 Generic Workflow Systems

Workflow management tools are key enablers of sustainable data analysis, which involves ensuring that computational workflows are reproducible, reusable, and understandable over time [15].

Workflow management tools such as Snakemake, the Common Workflow Language (CWL), and Nextflow provide structured frameworks for building and sharing data analysis pipelines. Snakemake allows researchers to explicitly define steps and dependencies, enabling clear documentation and reproducibility [15]. Similarly, CWL helps describe workflows in a platform-neutral format that enables interoperability and collaborative development [1]. Nextflow extends these ideas with strong support for scalable execution in cloud and high-performance computing environments [5]. Due to their generic nature, these systems share some limitations. They generally assume technical users are comfortable with writing workflow definitions in code or markup, which limits accessibility for clinicians and domain experts. User-facing interfaces remain limited, and integration with mobile and wearable platforms is not a primary design goal.

2.3 Workflow Applications in Digital Health

Tools like RAPIDS illustrate how workflow systems can be applied within digital health research. RAPIDS leverages Snakemake to enforce reproducibility and transparency in analytical processes [15]. Adopting a structured, workflow-based model enables better documentation of data analysis steps and promotes reuse. However, RAPIDS assumes users have technical expertise and programming skills, limiting accessibility for clinicians or domain experts. In addition, it primarily targets specific sensor data types and does not natively support mobile or wearable platforms.

In contrast, the Digital Biomarker Discovery Pipeline takes a notebook-based approach, providing reusable Jupyter notebooks¹ as templates for common analytical procedures [4]. This lowers barriers for entry, offering an interactive environment that is more familiar to some researchers. However, notebook-based pipelines have their own challenges; they lack modularity, are difficult to version, and do not natively manage software dependencies or execution environments, making reproducibility across systems a manual process.

¹<https://jupyter.org/>

3 Methodology

This research adopts a design science methodology, grounded in the systematic development and evaluation of an artefact—in this case, the CARP-DSP framework. Design science is well suited to this work, as it emphasizes both the creation of innovative artefacts and their rigorous evaluation in real-world contexts [17]. My approach follows three main phases: Requirements Analysis and Implementation, followed by Evaluation, enabling iterative refinement of the framework through cycles of design, evaluation, and stakeholder feedback.

3.1 Requirements Analysis

The first phase involved a comprehensive requirements analysis. This was informed by a critical review of digital phenotyping literature, an evaluation of existing workflow and data science frameworks, and discussions with researchers actively using the CARP platform. Based on this process, a series of researcher-driven scenarios were identified to reflect real-world needs and guide system design. As with the initial development of CARP, these scenarios served as the foundation for a domain-driven design process.

The identified scenarios span a broad range of use cases, from simple data export pipelines and quality assurance checks to complex, privacy-preserving edge/cloud analytics workflows and adaptive analytics workflows. In addition to the requirements emerging from these scenarios, the domain model explicitly incorporates data and processes necessary to support FAIR principles and sustainable research practices, ensuring that these are treated as first-class requirements rather than secondary concerns. The requirements further emphasize key system qualities such as reproducibility, adaptability, traceability, and metadata-rich documentation of workflows, directly aligning with the research questions driving this project.

A further guiding constraint was that the sub-system should conform to existing CARP design patterns, including a strong adherence to object-oriented principles and the domain-driven architecture of CARP, which separates core domain logic from infrastructural and application-specific components [6].

The requirements analysis has undergone several iterations, resulting in a detailed Requirements Analysis Document (RAD). This document currently informs the ongoing implementation and evaluation of the CARP-DSP framework. While not yet published, it is planned for release in the public CARP code repository.

3.2 Implementation

The framework is implemented as an extensible, workflow-based analytics module within the CARP ecosystem, known as CARP-DSP. It enables the authoring, deployment, and execution of human-readable data science workflows (DsWFs) spanning mobile devices, backend services, and hybrid configurations. The architecture emphasizes modularity, clear separation of concerns, and strong support for traceability and versioning of workflows.

A key distinction of CARP-DSP is that workflows are tightly integrated with study protocols within the CARP platform, ensuring that analytics pipelines are reproducible, linked to study metadata, and governed by the same versioning and deployment lifecycle as data collection protocols. In contrast to general-purpose

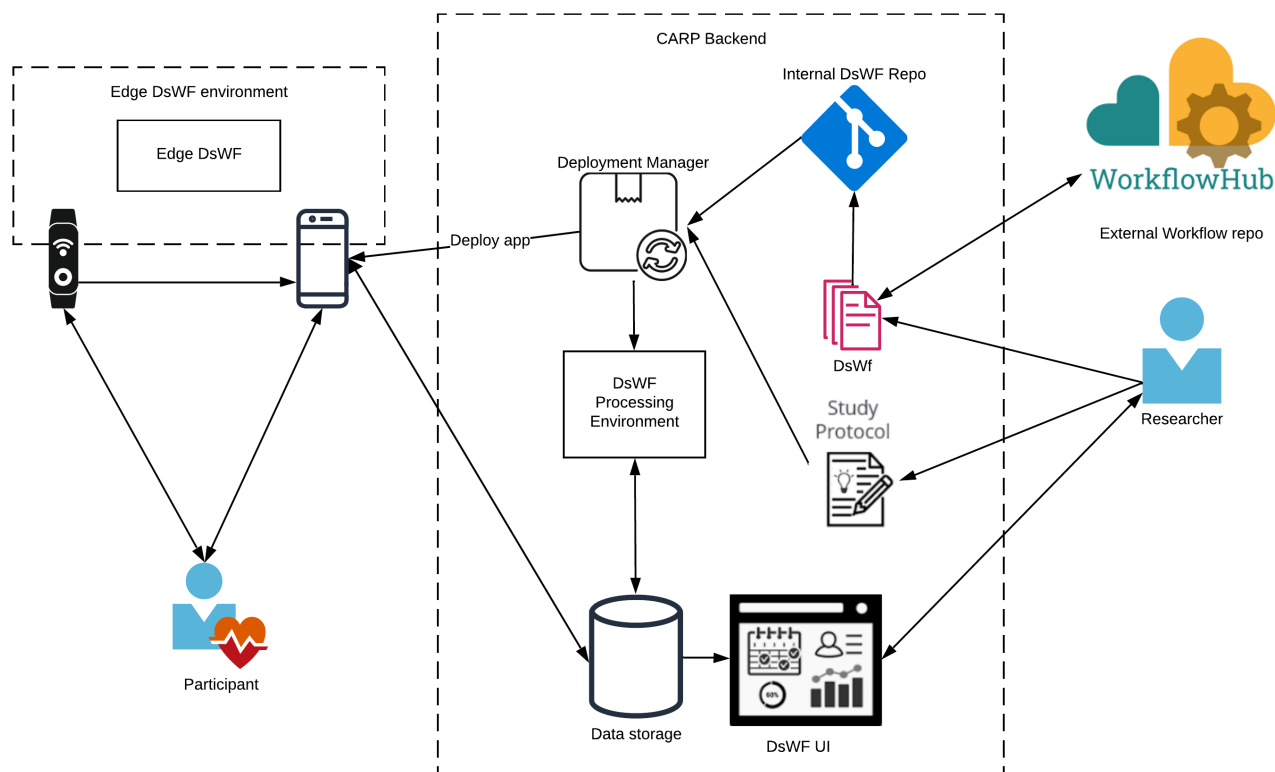


Figure 1: CARP-DSP system overview.

workflow engines such as Snakemake [15], Nextflow [5], or domain-specific tools like RAPIDS [25], CARP-DSP is designed specifically for mobile health and digital phenotyping contexts, with first-class support for FAIR principles, privacy-aware workflows, and cross-device execution. Related platforms such as DBDP [4], MindLAMP [23], and StudentLife [26] have shown the value of mobile sensing and digital biomarker workflows, but typically focus on specific data modalities or lack integrated workflow support for full life cycle study protocols and cross device analytics.

Figure 1 illustrates the high-level architecture of CARP-DSP. Researchers author DsWFs using a human-readable YAML-based DSL supported by a modular typing system and data descriptors. Workflows can be sourced from internal repositories or external resources such as WorkflowHub², facilitating workflow reuse and cross-study interoperability. To support cross-system studies, CARP-DSP will introduce a CWL translation layer, enabling workflows authored in other frameworks (e.g., AWARE’s RAPIDS) to be imported and executed within CARP-DSP, and conversely, allowing CARP-defined workflows to interoperate with external systems.

Each workflow is versioned and linked to a specific study protocol in the CARP backend, enabling rigorous traceability and reproducibility. The current implementation includes a custom-built

workflow execution engine, supporting batch and triggered execution models. The Trigger Service, currently in development, will enable flexible execution based on time schedules, data events, or study-defined conditions. The architecture supports template-based workflow reuse, allowing validated workflows to be repurposed across studies and institutions.

While the initial release of CARP-DSP focuses on backend-based workflow execution, the architecture is designed to support edge-based analytics in future releases. Planned enhancements include the introduction of a lightweight mobile interpreter for on-device workflow execution, with authoring tools that will evolve to support less verbose DSL and AI-assisted workflow authoring through LLM-based interfaces. Data lineage tracking, execution metadata, and workflow results are integrated through a user-facing DsWF UI, promoting transparency, reproducibility, and alignment with emerging FAIR practices.

4 Evaluation

The evaluation of the CARP-DSP framework will be grounded in practical studies and scenario-based testing, aligned with both the project's Research Questions and the use cases identified in the Requirements Analysis.

My research will aim to conduct case studies in collaboration with clinical researchers, targeting domains such as cardiovascular

²<https://workflowhub.eu>

health, stress monitoring, and mental health. These studies serve both to validate CARP-DSP's technical capabilities and to assess its usability and integration within existing research workflows. Particular emphasis is placed on supporting FAIR-aligned analytics capabilities.

Usability is assessed through user studies involving clinical researchers and data scientists, focusing on workflow authoring, transparency, and interpretability. Early feedback informs iterative refinement of the DSL and authoring tools.

Reproducibility is evaluated by replicating analyses across multiple datasets, devices, and deployment environments, testing CARP-DSP's ability to maintain consistent and robust outputs in heterogeneous mobile health settings.

Finally, comparative analyses will also be considered to assess the benefits of the CARP-DSP approach relative to baseline practices in digital phenotyping, such as Jupyter-based pipelines, ad-hoc analysis scripts, or platform-specific workflows. Key evaluation criteria include workflow reusability, efficiency, transparency, and reproducibility gains.

5 (Expected) Contribution

This research will contribute a workflow-based analytics framework specifically tailored for mobile health and digital phenotyping contexts. By integrating structured workflows tightly with study protocols, CARP-DSP will enable reproducible, FAIR-aligned, and privacy-aware data science pipelines spanning backend and edge environments.

Beyond the technical artefact, the project will provide concrete demonstrations of how structured workflows can improve transparency, adaptability, and reproducibility in mHealth research. The project also aims to contribute to emerging community efforts around workflow sharing and interoperability, through planned support for CWL translation and integration with community resources such as WorkflowHub.

More broadly, CARP-DSP is intended to democratize the authoring and reuse of digital phenotyping workflows, enabling a wider range of researchers, including clinicians and domain experts, to participate in the design, deployment, and interpretation of data science pipelines.

References

- [1] Peter Amstutz, Michael R. Crusoe, Nebojša Tijanić, Brad Chapman, John Chilton, Michael Heuer, Andrey Kartashov, Dan Leehr, Hervé Ménager, Maya Nedeljkovich, Matt Scales, Stian Soiland-Reyes, and Luka Stojanovic. 2016. Common Workflow Language, v1.0. (2016). doi:10.6084/m9.figshare.3115156.v2
- [2] Jakob E. Bardram. 2020. *The CARP Mobile Sensing Framework – A Cross-platform, Reactive, Programming Framework and Runtime Environment for Digital Phenotyping*. doi:10.48550/arXiv.2006.11904 arXiv:2006.11904 [cs]
- [3] Jakob E. Bardram, Claus Cramer-Petersen, Alban Maxhuni, Mads V.S. Christensen, Per Bækgaard, Dan R. Persson, Nanna Lind, Merete B. Christensen, Kirsten Nørgaard, Jayden Khakurel, Timothy C. Skinner, Dagmar Kownatka, and Allan Jones. 2023. DiaFocus: A Personal Health Technology for Adaptive Assessment in Long-Term Management of Type 2 Diabetes. 4, 2 (2023). doi:10.1145/3586579
- [4] Brinnae Bent, Ke Wang, Emilia Grzesiak, Chentian Jiang, Yuankai Qi, Yihang Jiang, Peter Cho, Kyle Zingler, Felix Ikponmwoosa Ogbode, Arthur Zhao, Ryan Runge, Ida Sim, and Jessilyn Dunn. 2020. The Digital Biomarker Discovery Pipeline: An Open-Source Software Platform for the Development of Digital Biomarkers Using mHealth and Wearables Data. 5, 1 (2020), e19. doi:10.1017/cts.2020.511 pubmed:33948242
- [5] Paolo Di Tommaso, Maria Chatzou, Evan W. Floden, Pablo Prieto Barja, Emilio Palumbo, and Cedric Notredame. 2017. Nextflow Enables Reproducible Computational Workflows. 35, 4 (2017), 316–319. doi:10.1038/nbt.3820
- [6] Eric Evans. 2004. *Domain-driven design: tackling complexity in the heart of software*. Addison-Wesley Professional.
- [7] M. Faurholt-Jepsen, J. Busk, M. Frost, M. Vinberg, E. M. Christensen, O. Winther, J. E. Bardram, and L. V. Kessing. 2016. Voice Analysis as an Objective State Marker in Bipolar Disorder. 6, 7 (2016), e856–e856. doi:10.1038/tp.2016.123
- [8] Denzil Ferreira, Vassilis Kostakos, and Anind K. Dey. 2015. AWARE: Mobile Context Instrumentation Framework. 2 (2015). https://www.frontiersin.org/articles/10.3389/fict.2015.00006
- [9] Kit Huckvale, Svetha Venkatesh, and Helen Christensen. 2019. Toward Clinical Digital Phenotyping: A Timely Opportunity to Consider Purpose, Quality, and Safety. 2, 1 (2019), 1–11. doi:10.1038/s41746-019-0166-1
- [10] Utkarshani Jaimini, Krishnaprasad Thirunarayan, Maninder Kalra, Revathy Venkataraman, Dipesh Kadariya, and Amit Sheth. 2018. "How Is My Child's Asthma?" Digital Phenotype and Actionable Insights for Pediatric Asthma. 1, 2 (2018), e11988. doi:10.2196/11988
- [11] Sachin H. Jain, Brian W. Powers, Jared B. Hawkins, and John S. Brownstein. 2015. The Digital Phenotype. 33, 5 (2015), 462–463. doi:10.1038/nbt.3223
- [12] Michael R. Jennings, Colin Turner, Raymond R. Bond, Alan Kennedy, Ranul Thantilage, Mohand Tahar Kechadi, Nhien-An Le-Khac, James McLaughlin, and Dewar D. Finlay. 2021. Code-Free Cloud Computing Service to Facilitate Rapid Biomedical Digital Signal Processing and Algorithm Development. *Computer Methods and Programs in Biomedicine* 211 (Nov. 2021), 106398. doi:10.1016/j.cmpb.2021.106398
- [13] Jasper S. Lee, Emma Browning, Joanne Hokayem, Hannah Albrecht, Georgia R. Goodman, Krishna Venkatasubramanian, Arlen Dumas, Stephanie P. Carreiro, Conall O'Cleirigh, and Peter R. Chai. 2024. Smartphone and Wearable Device-Based Digital Phenotyping to Understand Substance Use and Its Syndemics. 20, 2 (2024), 205–214. doi:10.1007/s13181-024-01000-5 pubmed:38436819
- [14] Francisco Martinez-Lasaca, Pablo Diez, Esther Guerra, and Juan de Lara. [n. d.]. LowcoBot: Towards Chatting with Low-Code Platforms. ([n. d.]).
- [15] Felix Mölder, Kim Philipp Jablonski, Brice Letcher, Michael B. Hall, Christopher H. Tomkins-Tinch, Vanessa Sochat, Jan Forster, Soohyun Lee, Sven O. Twardziok, Alexander Kanitz, Andreas Wilm, Manuel Holtgrewe, Sven Rahmann, Sven Nahnsen, and Johannes Köster. 2021. Sustainable Data Analysis with Snakemake. 10:33 (2021). doi:10.12688/f1000research.29032.1
- [16] Jukka-Pekka Onnela, Caleb Dixon, Keary Griffin, Tucker Janicke, Leila Minnowada, Sean Esterkin, Alvin Siu, Josh Zagorsky, and Eli Jones. 2021. Beiwe: A Data Collection Platform for High-Throughput Digital Phenotyping. 6, 68 (2021), 3417. doi:10.21105/joss.03417
- [17] Ken Peffers, Tuunanen, Tuure, Rothenberger, Marcus A., and prefix=and useprefix=true family=Chatterjee, given=Samir. 2007. A Design Science Research Methodology for Information Systems Research. 24, 3 (2007), 45–77. doi:10.2753/MIS0742-1222240302
- [18] Yatharth Ranjan, Zulqarnain Rashid, Callum Stewart, Pauline Conde, Mark Begale, Denny Verbeek, Sebastian Boettcher, Richard Dobson, and Amos Folarin. 2019. RADAR-Base: Open Source Mobile Health Platform for Collecting, Monitoring, and Analyzing Data Using Sensors, Wearables, and Mobile Devices. 7, 8 (2019), e11734. doi:10.2196/11734
- [19] Sohrab Saeb, Emily G. Lattie, Stephen M. Schueller, Konrad P. Kording, and David C. Mohr. 2016. The Relationship between Mobile Phone Location Sensor Data and Depressive Symptom Severity. 4 (2016), e2537. doi:10.7717/peerj.2537
- [20] Stephen M. Schueller, Mark Begale, Frank J. Penedo, and David C. Mohr. 2014. Purple: A Modular System for Developing and Deploying Behavioral Intervention Technologies. *Journal of Medical Internet Research* 16, 7 (July 2014), e3376. doi:10.2196/jmir.3376
- [21] Jing Xian Teo, Sonia Davila, Chengxi Yang, An An Hii, Chee Jian Pua, Jonathan Yap, Swee Yaw Tan, Anders Sahlén, Calvin Woon-Loong Chin, Bin Tean Teh, Steven G. Rozen, Stuart Alexander Cook, Khung Keong Yeo, Patrick Tan, and Weng Khong Lim. 2019. Digital Phenotyping by Consumer Wearables Identifies Sleep-Associated Markers of Cardiovascular Disease Risk and Biological Aging. 2 (2019), 361. doi:10.1038/s42003-019-0605-1 pubmed:31602410
- [22] John Torous, Mathew V. Kiang, Jeanette Lorme, and Jukka-Pekka Onnela. 2016. New Tools for New Research in Psychiatry: A Scalable and Customizable Platform to Empower Data Driven Smartphone Research. 3, 2 (2016), e5165. doi:10.2196/mental.5165
- [23] Aditya Vaidyam, John Halamka, and John Torous. 2022. Enabling Research and Clinical Use of Patient-Generated Health Data (the mindLAMP Platform): Digital Phenotyping Study. 10, 1 (2022), e30557. doi:10.2196/30557
- [24] Niels Van Berckel, Simon D'Alfonso, Rio Kurnia Susanto, Denzil Ferreira, and Vassilis Kostakos. 2023. AWARE-Light: A Smartphone Tool for Experience Sampling and Digital Phenotyping. *Personal and Ubiquitous Computing* 27, 2 (April 2023), 435–445. doi:10.1007/s00779-022-01697-7
- [25] Julio Vega, Meng Li, Kwesi Aguilera, Nikunj Goel, Echhit Joshi, Kirtiraj Khandekar, Krina C. Durica, Abhineth R. Kunta, and Carissa A. Low. 2021. Reproducible Analysis Pipeline for Data Streams: Open-Source Software to Process Data Collected With Mobile Devices. 3 (2021). doi:10.3389/fdgth.2021.769823
- [26] Rui Wang, Fanglin Chen, Zhenyu Chen, Tianxing Li, Gabriella Harari, Stefanie Tignor, Xia Zhou, Dror Ben-Zeev, and Andrew T. Campbell. 2014. StudentLife:

- Assessing Mental Health, Academic Performance and Behavioral Trends of College Students Using Smartphones. In *Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Seattle Washington, 2014-09-13). ACM, 3–14. doi:10.1145/2632048.2632054
- [27] Xuancong Wang, Nikola Vouk, Creighton Heaukulani, Thisum Buddhika, Wijaya Martanto, Jimmy Lee, and Robert Jt Morris. 2021. HOPES: An Integrative Digital Phenotyping Platform for Data Collection, Monitoring, and Machine Learning. 23, 3 (2021), e23984. doi:10.2196/23984
- [28] Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C. 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao, and Barend Mons. 2016. The FAIR Guiding Principles for Scientific Data Management and Stewardship. 3, 1 (2016), 160018. doi:10.1038/sdata.2016.18
- [29] Gent Ymeri, Nigel Sjölin Grech, Myrthe Wassenburg, Carl Magnus Olsson, Per Svenningsson, and Dario Salvi. 2024. Measuring Finger Dexterity in Parkinson's Disease with Mobile Phones. In *2024 IEEE International Conference on Pervasive Computing and Communications Workshops and Other Affiliated Events (PerCom Workshops)* (2024-03). 112–116. doi:10.1109/PerComWorkshops59983.2024.10503245